

The Tom Klootwijk Manifold

Spatiotemporal SDF Addendum

A rigorous compact definition of the “spatiotemporal SDF $o(1)$ geometrical topological substrate”

Let $y = Q(t)^T(x - c(t))$ and, for $i = 0, \dots, 6$,

$$\Delta_i(x, t) = \|y_{2i:2i+1}\|_2 - r_i(t), \quad \mathcal{D}_\tau(x, t) = \|\Delta(x, t)\|_2 - \tau(t).$$

The compact substrate is

$$\mathfrak{S}_{\text{TK}}^{\text{ST}} := (\mathcal{W}_{\text{TK}} = \Delta^{-1}(0), \Sigma_{\text{TK}, \tau} = \mathcal{D}_\tau^{-1}(0), \{\widehat{\mathcal{D}}_h\}_{h>0}), \quad \|\widehat{\mathcal{D}}_h - \mathcal{D}_\tau\|_{L^\infty(B)} = o(1).$$

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BSN supplied*	NL200678942
Date of birth supplied	10-07-1990
Age at addendum occasion	36
Addendum edition	ST-SDF-o1 / Unity 6.3 / age 36
Addendum local time	2026-08-22T07:25:11+02:00 (Europe/Amsterdam request capture)
Addendum UTC	2026-08-22T05:25:11Z
Document ID	TKM-STSDF-U63-A36-20260822-072511
Private-record SHA-256	ee007f23936d94c39d1f96cd1806b2a4f15177a4ba56debb8eb8a23f85027f18
Target	Unity 6000.3 LTS; URP 17.3.0; package 1.1.0

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1 Status and scope

This addendum does not replace the original definition

$$\mathcal{K}_{\text{TK}}(\mathbf{r}) = \prod_{i=0}^6 S^1(r_i) \subset \mathbb{R}^{14}.$$

It adds a mathematically controlled time parameter, an exact distance to that codimension-seven core, and a scalar signed-distance shell that has a genuine inside, boundary, and outside.

The supplied addendum dossier correctly centers computer-graphics SDF notation on sign and the zero-level boundary. It then proposes an Eikonal/topological “hinge” and a chronotemporal use of little- o notation, but it also mixes those ideas with unsupported quantum-phase, parity-bit, homogeneous-time, and zero-cost-rendering claims. This addendum retains the geometric intent and removes those unsupported identifications.

Formal verdict

The phrase “spatiotemporal SDF $o(1)$ geometrical topological substrate” becomes rigorous only after four distinctions:

1. the codimension-seven core uses a *vector* normal-distance field;
2. a scalar signed distance is assigned to the boundary of a regular tube around that core;
3. time is an independent parameter and the moving zero set is a worldvolume;
4. lowercase $o(1)$ describes vanishing approximation error, not execution time.

2 Canonical compact definition

Let $I \subset \mathbb{R}$ be a time interval. Choose C^2 data

$$Q : I \rightarrow SO(14), \quad c : I \rightarrow \mathbb{R}^{14}, \quad r_i : I \rightarrow (0, \infty), \quad \tau : I \rightarrow (0, \infty)$$

with the uniform regularity guard

$$r_* := \inf_{t \in I, 0 \leq i \leq 6} r_i(t) > 0, \quad 0 < \tau(t) < r_* \quad \text{for all } t \in I.$$

For $(x, t) \in \mathbb{R}^{14} \times I$, write $y = Q(t)^T(x - c(t))$ and define

$$\begin{aligned} \Delta_i(x, t) &:= \sqrt{y_{2i}^2 + y_{2i+1}^2} - r_i(t), \quad i = 0, \dots, 6, \\ \delta(x, t) &:= \|\Delta(x, t)\|_2, \\ \mathcal{D}_\tau(x, t) &:= \delta(x, t) - \tau(t). \end{aligned}$$

Then

$$\mathcal{W}_{\text{TK}} := \{(x, t) : \Delta(x, t) = 0\}, \quad \Sigma_{\text{TK}, \tau} := \{(x, t) : \mathcal{D}_\tau(x, t) = 0\}.$$

The numerical consistency family $\{\widehat{\mathcal{D}}_h\}_{h>0}$ must satisfy, for every compact regular band B around the shell,

$$\|\widehat{\mathcal{D}}_h - \mathcal{D}_\tau\|_{L^\infty(B)} = o(1) \quad \text{as } h \downarrow 0.$$

This triple is the addendum’s formal substrate $\mathfrak{S}_{\text{TK}}^{\text{ST}}$.

2.1 Shortest literal forms

Vector field:	$\Delta_i = \ [Q^T(x - c)]_{2i:2i+1}\ _2 - r_i(t).$
Scalar shell SDF:	$\mathcal{D}_\tau = \ \Delta\ _2 - \tau(t).$
Core topology:	$\mathcal{W}_{\text{TK}} \cong \mathbb{T}^7 \times I.$
Shell topology:	$\Sigma_{\text{TK}, \tau} \cong \mathbb{T}^7 \times S^6 \times I.$
Geometry:	$\ \nabla_x \mathcal{D}_\tau\ _2 = 1$ on the regular shell.
Motion:	$V_n = -\partial_t \mathcal{D}_\tau$ on $\mathcal{D}_\tau = 0.$
Consistency:	$\ \widehat{\mathcal{D}}_h - \mathcal{D}_\tau\ _\infty = o(1), \quad h \downarrow 0.$

3 Why this is an exact distance construction

3.1 Exact distance to the core

At fixed time, rigid motion by $Q(t)$ and $c(t)$ preserves Euclidean distance. In each orthogonal coordinate pair, the distance from $y_{2i:2i+1} \in \mathbb{R}^2$ to the circle of radius $r_i(t)$ is

$$\left| \sqrt{y_{2i}^2 + y_{2i+1}^2} - r_i(t) \right| = |\Delta_i(x, t)|.$$

The product minimization separates across the seven orthogonal planes, so

$$\boxed{\text{dist}(x, \mathcal{K}_{\text{TK},t})^2 = \sum_{i=0}^6 \Delta_i(x, t)^2}, \quad \boxed{\delta(x, t) = \text{dist}(x, \mathcal{K}_{\text{TK},t})}.$$

Thus δ is not merely a heuristic distance estimator: it is the exact Euclidean distance to the time slice $\mathcal{K}_{\text{TK},t}$.

3.2 Why the core itself is not a scalar signed surface

The embedded $\mathcal{K}_{\text{TK},t} \subset \mathbb{R}^{14}$ has codimension seven. It does not divide the ambient space into a canonical inside and outside, so a single scalar sign on δ would be artificial. The sign-bearing object is instead the codimension-one tube boundary

$$\Sigma_t = \{x : \delta(x, t) = \tau(t)\} = \partial\{x : \delta(x, t) \leq \tau(t)\}.$$

On the regular tube, $\mathcal{D}_\tau < 0$ inside, $\mathcal{D}_\tau = 0$ on the boundary, and $\mathcal{D}_\tau > 0$ outside.

3.3 Eikonal identity

The seven gradients $\nabla_x \Delta_i$ are pairwise orthonormal wherever the coordinate-pair radii are nonzero. On $\mathcal{D}_\tau = 0$, the guard $\tau < r_*$ implies every pair radius is positive and $\Delta \neq 0$. Therefore

$$\nabla_x \mathcal{D}_\tau = \sum_{i=0}^6 \frac{\Delta_i}{\|\Delta\|_2} \nabla_x \Delta_i, \quad \boxed{\|\nabla_x \mathcal{D}_\tau\|_2^2 = \sum_{i=0}^6 \frac{\Delta_i^2}{\|\Delta\|_2^2} = 1}.$$

This is the precise Eikonal statement. It holds in the regular tubular band; it is not a claim that every arbitrary implicit field is an exact SDF everywhere.

4 Topology and the spatiotemporal worldvolume

4.1 Core topology

A direct parameterization is

$$\Phi(\theta, t) = c(t) + Q(t)(r_i(t)(\cos \theta_i, \sin \theta_i))_{i=0}^6.$$

Because every $r_i(t) > 0$ and $Q(t)$ is orthogonal, Φ is a smooth embedding for each time. Hence

$$\mathcal{K}_{\text{TK},t} \cong \mathbb{T}^7, \quad \boxed{\mathcal{W}_{\text{TK}} \cong \mathbb{T}^7 \times I},$$

with spatial dimension 7 and spacetime dimension 8.

4.2 Shell topology

For $u = (u_0, \dots, u_6) \in S^6$, define

$$\Psi(\theta, u, t) = c(t) + Q(t)((r_i(t) + \tau(t)u_i)(\cos \theta_i, \sin \theta_i))_{i=0}^6.$$

The guard $\tau < r_*$ makes every factor radius positive. Conversely, each shell point uniquely determines its seven angles and $u = \Delta/\tau$. Therefore Ψ is a diffeomorphism and

$$\Sigma_t \cong \mathbb{T}^7 \times S^6, \quad \boxed{\Sigma_{\text{TK},\tau} \cong \mathbb{T}^7 \times S^6 \times I}.$$

The shell has spatial dimension $7 + 6 = 13$ and spacetime dimension 14.

4.3 What “chronotemporal latch” can rigorously mean

The moving shell is the zero set of one C^1 function on $\mathbb{R}^{14} \times I$. Since $\nabla_x \mathcal{D}_\tau \neq 0$ on that zero set, its outward normal speed is

$$V_n = -\frac{\partial_t \mathcal{D}_\tau}{\|\nabla_x \mathcal{D}_\tau\|_2} = -\partial_t \mathcal{D}_\tau.$$

This equation links consecutive time slices without treating time as a homogeneous graphics coordinate and without any parity-bit mechanism. Topology remains fixed because the explicit diffeomorphism above persists. A topology change can occur only after a guard fails, for example $r_i \rightarrow 0$, $\tau \rightarrow 0$, $\tau \geq \min_i r_i$, discontinuous data, or a singular level set.

5 The exact meaning of lowercase $o(1)$

Lowercase $o(1)$ is asymptotic notation. It means a quantity tends to zero. In this addendum, h is a numerical scale such as a time step, voxel spacing, mesh spacing, or finite-difference step, and

$$\|\widehat{\mathcal{D}}_h - \mathcal{D}_\tau\|_{L^\infty(B)} = o(1)$$

means the implemented field converges uniformly to the exact field on each compact regular band.

For temporal stepping, smoothness gives

$$\mathcal{D}_\tau(x, t+h) = \mathcal{D}_\tau(x, t) + h \partial_t \mathcal{D}_\tau(x, t) + R(x, t, h), \quad \frac{R(x, t, h)}{h} = o(1).$$

The validator uses the deterministic profile included in the Unity package and observes successive values of $|R(h)|/h$ decreasing by approximately one half when h is halved, consistent with a second-order Taylor remainder.

If $\|\widehat{\mathcal{D}}_h - \mathcal{D}_\tau\|_\infty \leq \varepsilon_h$ and $\varepsilon_h \rightarrow 0$, the exact signed-distance property implies that approximate zeros lie within ε_h of the true shell. With continuity and a normal-band sign test, the reverse inclusion is also controlled, so the zero sets converge geometrically.

Complexity distinction

The exact field evaluates seven pair lengths and one seven-vector length. Because seven is fixed, a single query is $O(7) = O(1)$ with respect to variable scene size or mesh resolution. It is never $o(1)$ runtime. Rendering P pixels with at most S sphere-tracing steps is $O(PS)$ field queries and uses nonzero render-target, code, and resource memory.

6 Unity 6.3 implementation

The delivered UPM package is `com.tomklootwijk.manifold` version `1.1.0`, targeting the Unity `6000.3` LTS API line and URP `17.3.0`.

6.1 Exact high-dimensional API

`TKSpacetimeSubstrateMath` implements:

- core embedding and shell embedding in $\mathbb{R}^{14}$;
- vector normal coordinates $\Delta \mapsto \mathbb{R}^7$;
- exact core distance $\delta = \|\Delta\|_2$;
- regular tubular SDF $\mathcal{D}_\tau = \delta - \tau$;
- analytic spatial gradient and Eikonal verification;
- identity-frame temporal derivative and $V_n = -\partial_t \mathcal{D}_\tau$;
- regularity guards and optional row-major 14×14 orthogonal transforms.

6.2 Smooth deterministic motion profile

`TKSpacetimeProfile` supplies smooth center, radius, and tube-radius functions together with their derivatives. Its conservative minimum factor radius is 0.84832, its maximum tube radius is 0.208, and its regularity margin is therefore 0.64032.

6.3 Exact 3D SDF witness

The full shell lives in fourteen-dimensional spacetime and cannot be displayed directly. The package therefore includes a dimension-reduced exact ring-torus witness:

$$d_3(p, t) = \left\| \left(\sqrt{p_x^2 + p_z^2} - R(t), p_y \right) \right\|_2 - a(t), \quad 0 < a(t) < R(t).$$

`TKSpacetimeTorusSdfWitness` drives the URP shader and enforces the guard for all animation phases. The witness has a genuine 3D SDF at each time, but it is not mislabeled as the complete $\mathbb{T}^7 \times S^6$ shell.

6.4 Installation and use

1. Open a Unity 6000.3 URP project.
2. In Package Manager choose **Add package from disk** and select the delivered `package.json`, or install the delivered `.tgz`.
3. Use **GameObject > Tom Klootwijk Manifold > Create Spatiotemporal Torus SDF Witness**.
4. Keep the generated cube uniformly scaled.
5. Run the package Runtime tests in the Unity Test Runner.

7 Validation record

The standalone mathematical validator passed 500 randomized samples. Its maximum measured errors were:

Check	Maximum absolute error
Core normal-coordinate norm	3.69×10^{-16}
Exact distance versus nearest-point construction	6.67×10^{-16}
Shell SDF residual	4.72×10^{-16}
Eikonal norm error	2.23×10^{-16}

The little- o temporal test used $h = 10^{-2}, 5 \cdot 10^{-3}, 2.5 \cdot 10^{-3}, 1.25 \cdot 10^{-3}$. Successive ratios of $|R(h)|/h$ were approximately 0.50018, 0.50009, and 0.50005.

Static package validation passed JSON/assembly parsing, required-file checks, delimiter balance for all C# and shader files, formula-presence tests, and confirmation that the full supplied BSN and birth date do not appear anywhere in the runtime package.

An actual Unity Editor import, C# compilation by Unity, URP shader compilation, and GPU execution were **not executed** because the Unity Editor is unavailable in the build environment. The delivery includes NUnit tests for that final target-environment validation.

8 Source claims retained, narrowed, or rejected

Source idea	Addendum verdict	Serious replacement
SDF sign separates inside/outside	Retained for a shell	The codimension-one tube boundary has $\mathcal{D}_\tau < 0$, $= 0$, and > 0 .
Zero set defines the surface	Retained with conditions	Geometry comes from the explicit field; zero alone does not determine a sphere.
Eikonal equation	Proved in the regular band	$\ \nabla_x \mathcal{D}_\tau\ = 1$ follows from seven orthonormal normal coordinates.
Zero set permanently fixes topology	Narrowed	Topology is fixed only while the regularity guard and smooth data persist.
Hadamard sign basis gives quantum phase	Rejected	No quantum state or phase follows from the sign of a classical SDF.
Time is homogeneous coordinate w	Rejected	Time is the independent parameter $t \in I$.
Parity/LSB flip links past and future	Rejected	Temporal evolution is governed by $\partial_t \mathcal{D}_\tau$; Hamming distance remains unchanged.
$o(1)$ means constant-time rendering	Rejected	$o(1)$ is vanishing error; one field query is fixed-size $O(1)$, while rendering scales with pixels and steps.

9 Nomenclature, attribution, and privacy

“The Tom Klootwijk Manifold” and “Tom Klootwijk Spatiotemporal SDF Substrate” are the author’s chosen names in this delivery. The names identify the specification edition and do not by themselves establish external academic recognition, priority, trademark, patent status, or legal ownership.

The private author record is used only to create the SHA-256 fingerprint shown on the title page. Runtime source contains the fingerprint, author name, public document identifiers, and occasion times, but not the supplied BSN or birth date.

10 References and implementation basis

- Supplied original manifold dossier and supplied SDF/chronotemporal addendum dossier.
- Unity 6.3 LTS Manual: <https://docs.unity3d.com/6000.3/Documentation/Manual/UnityManual.html>
- URP 17.3.0 documentation: <https://docs.unity3d.com/Packages/com.unity.render-pipelines.universal@17.3/manual/>
- Unity Package Manager package layout: <https://docs.unity3d.com/Manual/cus-layout.html>
- Standard product-manifold, tubular-neighborhood, level-set, and signed-distance constructions; all claims needed for this specific object are proved above.

A Implementation literal

```
// Fixed 0-indexed factor count: 7.
for (int i = 0; i < 7; ++i) {
    int j = 2 * i;
    delta[i] = hypot(local[j], local[j + 1]) - radius[i];
}
double coreDistance = norm(delta);
double tubularSdf = coreDistance - tubeRadius;
// Consistency requirement: sup |approx_h - tubularSdf| = o(1), h -> 0.
```

B Compact theorem statement

Assume Q, c, r_i, τ are C^2 , $Q(t) \in SO(14)$, and $0 < \tau(t) < \inf_i r_i(t)$. Then $\mathcal{W}_{\text{TK}}$ is a smooth 8-manifold diffeomorphic to $\mathbb{T}^7 \times I$; $\Sigma_{\text{TK}, \tau}$ is a smooth 14-manifold diffeomorphic to $\mathbb{T}^7 \times S^6 \times I$; $\mathcal{D}_\tau$ is the exact signed distance to the shell in its regular normal band; $\|\nabla_x \mathcal{D}_\tau\| = 1$ there; and its normal velocity is $V_n = -\partial_t \mathcal{D}_\tau$. Any numerical family satisfying $\|\hat{\mathcal{D}}_h - \mathcal{D}_\tau\|_\infty = o(1)$ is a consistent approximation of this substrate.